CCE Assignment

Unit 4 Data structures

BST Deletion of a Node with one child :

1. problem statement

write a c++ program to delete a node having exactly one child from a Binary search Tree (BST) & display the updated tree. The program should search for the specified node, check whether it has exactly one child, connect its parent to its child, & delete the selected node without disturbing the BST property.

2. objectives

1. To understand the structure & properties of a Binary search Tree.
2. To implement the deletion operation for a node having exactly one child.
3. To understand the relationship between parent & child nodes.
4. To maintain the BST property after deletion.
5. To display the updated tree using inorder traversal.
6. To develop problem-solving & programming skills using c++.

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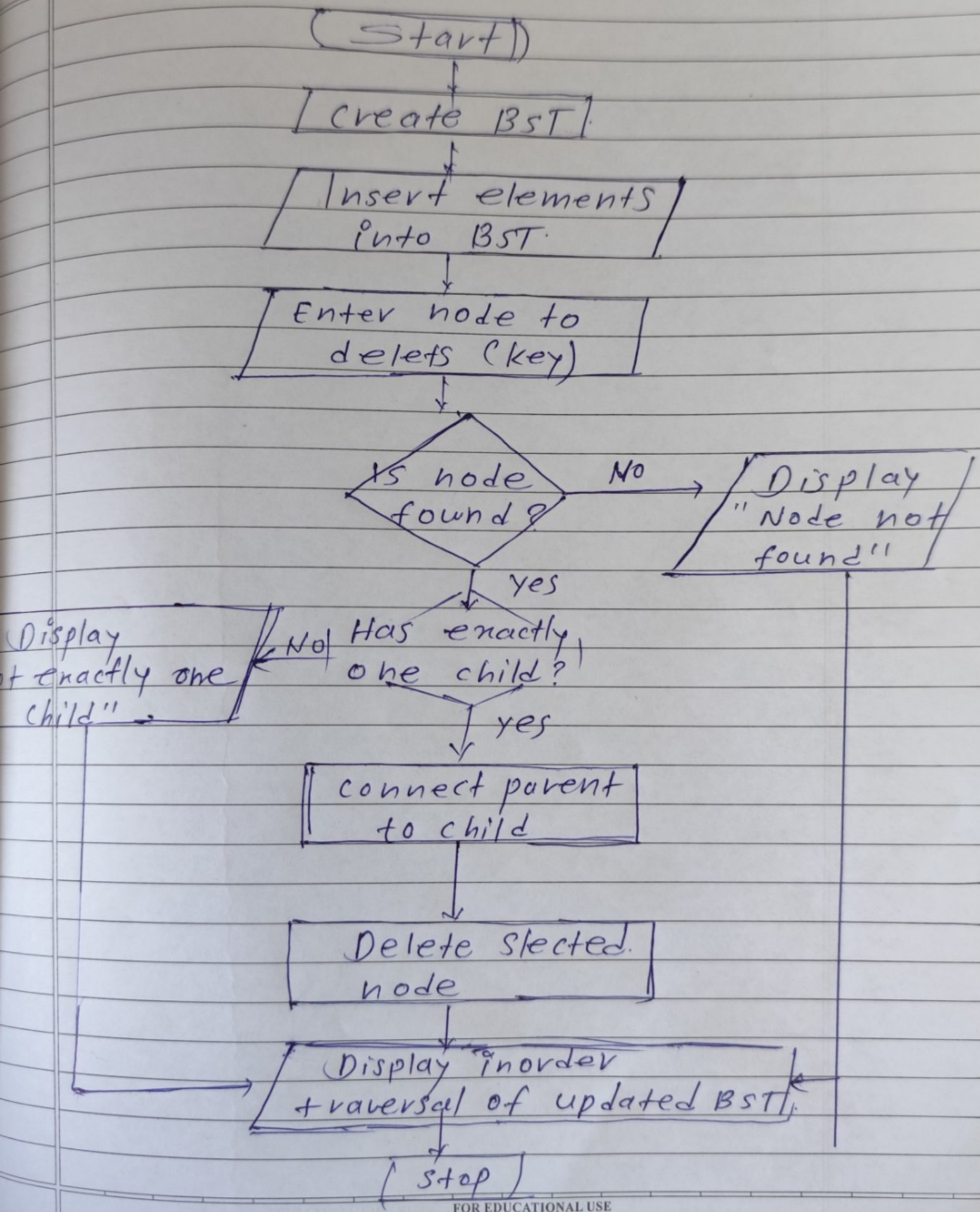
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3. Algorithm:

1. Start the program.
2. ~~Insert~~ create an empty Binary Search Tree.
3. Insert the given elements into the BST
4. Accept the key of the node to be deleted
5. Search for the node containing the specified key.
6. If the node is not found, display "Node not found".
7. If the node has exactly one child:
   - store the address of its child.
   - Connect the parent to that child.
   - Delete the selected node.
8. If the node does not have exactly one child, display an appropriate message.
9. Display the updated BST using inorder traversal.
10. stop the program.

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